

SAMY TAMIM

Software Engineer 1

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Experience

Software Engineer

June 2024 – Present

Mohammed VI Polytechnic University - UM6P

Rabat, Morocco

- Architected digital twin system for IoT monitoring with predictive maintenance capabilities, processing sensor data from industrial equipment through real-time analytics and MQTT integration
- Participated in Agile development with 2-week sprints, conducting weekly team meetings while collaborating with 1 backend developer on engineering best practices including code review and test-driven development
- Developed FastAPI REST services and Spring Boot microservices with PostgreSQL database, implementing OAuth2 authentication and exploring MySQL and MongoDB alternatives for scalable architecture
- Built Angular dashboards with TypeScript and implemented CI/CD pipelines using Docker and Jenkins, enabling stakeholders to identify maintenance issues proactively

Full-Stack Development Intern

November 2023 – May 2024

Mohammed VI Polytechnic University - UM6P

Rabat, Morocco

- Collaborated with cross-functional team to develop platform tracking 200+ daily ship movements for fertilizer trade, delivering real-time supply chain insights using Agile methodologies
- Engineered Flask REST API backend with MySQL database integration, optimizing data workflows with Pandas and improving dashboard performance through query optimization and caching strategies
- Developed interactive React.js maps with Leaflet.js for frontend visualization, implementing responsive design patterns for intuitive ship tracking interface

Projects

Real-time Multiplayer Platform | TypeScript, NestJS, Next.js, PostgreSQL, WebSockets, Docker

Sep 2023

- Built monolithic platform with Docker containerization, implementing event-driven architecture with WebSockets for real-time gameplay features
- Implemented OAuth2 authentication with 2FA and JWT session management following security best practices
- Contributed to team development using Agile practices, participating in code reviews and applying design patterns and refactoring techniques

Video Streaming Platform | Go, Next.js, PostgreSQL, Node.js, OAuth2

Dec 2025

- Built scalable Go backend with HLS streaming and optimized cache management for efficient video delivery
- Designed PostgreSQL schema with query optimization and connection pooling for improved data retrieval performance
- Collaborated with 3 developers (2 frontend, 1 backend) in Agile sprints, conducting sprint planning and daily standups for iterative delivery

Job Control System | Go, Async Jobs

Sep 2025

- Building job orchestration system with internal queue processing 1000+ async tasks per minute with +15 configuration options
- Implementing worker pool architecture with graceful shutdown, dead letter queues, and automatic retry mechanisms

Education

1337 Coding School - UM6P

November 2021 – Present

Software Engineering Program - Advanced Developer Badge (Level 16.09/21)

Khouribga, Morocco

- Pursuing Data & AI specialization track while completing peer-learning curriculum in algorithms, web development, and system programming

Technical Skills

Languages: TypeScript, JavaScript, Python, Go, Java, C, C++, OCaml, SQL

Frontend: React, Next.js, Angular, HTML/CSS, Tailwind CSS

Backend: Node.js, NestJS, Flask, FastAPI, Spring Boot, Express.js

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQL optimization

DevOps & Cloud: Docker, Jenkins, Git, AWS, CI/CD pipelines, Circle CI, Travis CI

Tools & Practices: WebSockets, REST APIs, OAuth2, JWT, MQTT, FFmpeg, Agile/Scrum, TDD, Code Review

Security & Architecture: OWASP Top 10, Microservices, Design Patterns, Test-Driven Development